

33–(MathAA/10_HL_Winter_2022_Q5)–ComplexNumbers

[Maximum mark: 7]

Consider the equation $z^3 + pz^2 + 54z^2 - 108z + 80 = 0$ where $z \in \mathbb{C}$ and $p \in \mathbb{R}$.

Three of the roots of the equation are $3 + i$, α and α^2 , where $\alpha \in \mathbb{R}$.

- (a) By considering the product of all the roots of the equation, find the value of α . [4]
- (b) Find the value of p . [3]



34-(MathAA/10_HL_Winter_2022_Q12)-ComplexNumbers

[Maximum mark: 18]

Let z_n be the complex number defined as $z_n = (n^2 + n + 1) + i$ for $n \in \mathbb{N}$.

- (a) (i) Find $\arg(z_n)$.
- (ii) Write down an expression for $\arg(z_n)$ in terms of n . [3]

Let $w_n = z_0 z_1 z_2 z_3 \dots z_{n-1} z_n$ for $n \in \mathbb{N}$.

- (b) (i) Show that $\arctan(a) + \arctan(b) = \arctan\left(\frac{a+b}{1-ab}\right)$ for $a, b \in \mathbb{R}$, $ab < 1$.
- (ii) Hence or otherwise, show that $\arg(w_1) = \arctan(2)$. [5]
- (c) Prove by mathematical induction that $\arg(w_n) = \arctan(n+1)$ for $n \in \mathbb{N}$. [10]



1-(MAT/10_HL_Winter_2014_Q10)-Permutation-Combination

[Maximum mark: 7]

A set of positive integers $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is used to form a pack of nine cards. Each card displays one positive integer without repetition from this set. Grace wishes to select four cards at random from this pack of nine cards.

- (a) Find the number of selections Grace could make if the largest integer drawn among the four cards is either a 5, a 6 or a 7. [3]
- (b) Find the number of selections Grace could make if at least two of the four integers drawn are even. [4]

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2-(MAT/10_HL_Winter_2018_Q2)-Permutation-Combination

A team of four is to be chosen from a group of four boys and four girls.

- (a) Find the number of different possible teams that could be chosen. [3]
- (b) Find the number of different possible teams that could be chosen, given that the team must include at least one girl and at least one boy. [2]



3-(MathAA/11_HL_Summer_2021_Q9)-Permutation-Combination

[Maximum mark: 8]

A farmer has six sheep pens, arranged in a grid with three rows and two columns as shown in the following diagram.

Five sheep called Amber, Brownie, Curly, Daisy and Eden are to be placed in the pens. Each pen is large enough to hold all of the sheep. Amber and Brownie are known to fight.

Find the number of ways of placing the sheep in the pens in each of the following cases:

- (a) Each pen is large enough to contain five sheep. Amber and Brownie must not be placed in the same pen. [4]
- (b) Each pen may only contain one sheep. Amber and Brownie must not be placed in pens which share a boundary. [4]



4-(MathAA/10_HL_Winter_2022_Q11)-Permutation-Combination,Remainder&FactorTheorem

[Maximum mark: 16]

Consider a three-digit code abc , where each of a , b and c is assigned one of the values 1, 2, 3, 4 or 5.

(a) Find the total number of possible codes

- (i) assuming that each value can be repeated (for example, 121 or 444);
- (ii) assuming that no value is repeated.

[4]

Let $P(x) = x^3 + ax^2 + bx + c$, where each of a , b and c is assigned one of the values 1, 2, 3, 4 or 5. Assume that no value is repeated.

Consider the case where $P(x)$ has a factor of $(x^2 + 3x + 2)$.

(b) (i) Find an expression for b in terms of a .

(ii) Hence show that the only way to assign the values is $a = 4$, $b = 5$ and $c = 2$.

(iii) Express $P(x)$ as a product of linear factors.

(iv) Hence or otherwise, sketch the graph of $y = P(x)$, clearly showing the coordinates of any intercepts with the axes.

[12]



1-(MAT/12_HL_Summer_2012_Q4)-BinomialTheorem

- (a) Expand and simplify $\left(x - \frac{2}{x}\right)^4$. [3 marks]
- (b) Hence determine the constant term in the expansion $(2x^2 + 1)\left(x - \frac{2}{x}\right)^4$. [2 marks]

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